Melvin Mathew

Data Scientist | AI/ML Engineer | Generative AI

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Summary

Data Scientist and AI/ML Engineer with hands-on experience in **Python**, **SQL**, **Machine Learning**, **Statistics**, **NLP** and **Generative AI**. Builds end-to-end **data pipelines** and models, from EDA and **feature engineering** to **deployment**: classification, regression and **time-series forecasting** with scikit-learn and XGBoost; deep learning in PyTorch and TensorFlow; and **LLM** applications with **RAG**, **embeddings**, vector databases, **prompt engineering** and AI agents. Ships with FastAPI REST APIs, Docker, CI/CD and GCP Cloud Run, backed by model monitoring, SHAP and **A/B testing**. Explains results to non-technical stakeholders. **Immediate joiner**.

Experience

Independent Data Scientist (Freelance) Jan 2026 – Present

Bengaluru. Freelance data science and AI work for clients, alongside portfolio projects; code public on GitHub.

Bike-Taxi Ride Demand Forecasting – *Clustering & time series forecasting, deployed on GCP* GitHub

- Built a **demand forecasting** pipeline on **8.38M** real ride bookings, clustering pickups into 50 geo-clusters.
- Modeled over-dispersed demand (37% zeros) with **XGBoost** on calendar and lag features, scored by Poisson deviance.
- Lag features took it from failing a seasonal-naive **deploy gate** (MASE 0.999) to passing on all 50 clusters (**0.804**).
- **Rolling-origin backtests**: under 5.2x demand growth a frozen model lost to naive by week 6, so set **4-weekly retraining**.
- **Deployed** it as a gated **FastAPI** service on **Cloud Run**, with a model registry, rollback and **352 tests**.

Nectar Intelligent Facilities Platform – *IoT analytics: predictive maintenance & anomaly detection* GitHub

- Built an IoT platform (XGBoost, NetworkX **graph analytics**) for **predictive maintenance** on 2M synthetic rows.
- Engineered an 82-feature, leakage-safe pipeline with forward-only targets, served through a live Streamlit app.
- Compared 4 **classifiers** by PR-AUC under class imbalance; Random Forest won (Precision 0.90, **ROC-AUC 0.896**).
- Built an XGBoost energy forecaster (**9.27% MAPE** vs 161% baseline) and an **anomaly-detection** framework.
- Proved reproducibility (retrain matched committed artifacts), backed by **55 tests** incl. no-lookahead leakage guards.

Data Scientist Sep 2025 – Dec 2025

upGrad School of Technology

Promoted from Data Science Content Strategist Associate after upGrad acquired Aimerz.ai – continued with the same team.

- Designed **SQL** and **Python** workflows to extract, clean and engineer features across structured and unstructured data.
- Built and tested **machine learning** and predictive analytics workflows with Pandas and Scikit-learn.
- Performed IQR-based anomaly detection, trend and correlation analysis to identify key business drivers.
- Developed **KPI dashboards** and business reports with Python, SQL, Pandas, NumPy, Matplotlib and Seaborn.
- Led the **AI Fluency** initiative, designing bootcamps on Prompt Engineering, AI Tools and **Generative AI**.

Data Science Content Strategist Associate Aug 2024 – Aug 2025

Aimerz.ai (acquired by upGrad in 2025)

- Joined as first member of Aimerz's Data Science team, building its first **15**+ Python ML workflows and projects.
- Automated data cleaning, **exploratory data analysis (EDA)** and feature engineering with Python and SQL.
- Applied regression, classification and statistical modelling, validating model performance.
- Designed data science curricula, labs and capstone projects, from Python foundations to Generative AI.

Projects

AML Fraud Detection Pipeline – *Production ML system, live on GCP Cloud Run* GitHub

- Deployed an end-to-end **ML pipeline** on **6.36M** transactions as a **FastAPI REST API** on Cloud Run at $0.
- Ran 3-fold time-based cross-validation across 8 models with Optuna and **MLflow**, selecting LightGBM on PR-AUC.
- Set the operating point from analyst capacity: **95.1% recall at zero false positives** at 250 reviews/day (500/day advised).
- Shipped with **Docker**, keyless GitHub Actions **CI/CD** (Workload Identity Federation) and structural rollback.
- Added **PSI drift monitoring** over 31 time windows and a daily cron on the live service; **234 tests**.

Graph-Orchestrated LLM Customer Support Agent – *Agentic AI, RAG & LLM evaluation* GitHub

- Built an **agentic AI** support assistant in **LangChain** as a node/edge conversation graph with tool calling.
- Grounded answers with **RAG** scoped to each user's subscription tier: **100% retrieval recall**, 0% cross-tier leakage.
- Tuned the answer prompt and knowledge base, cutting hallucination from **20% to 6.7%** (10% on held-out questions).
- Built a **237-conversation golden set** and a required CI eval gate on every PR, exposing and fixing 3 identity bypasses.
- Local-by-default on Ollama (Llama 3.2); OpenAI opt-in raised callback precision 94% to 98% for $0.03; **513 unit tests**.

Credit Default Risk Model – *Classification, explainability & model audit* GitHub

- Reproduced a **LightGBM** default model claiming 0.9636 AUC and proved **85% of its Gini was leakage**.
- Rebuilt it with walk-forward CV (hold-out Gini 0.2914), **SHAP** reason codes and isotonic calibration.
- Set a cutoff cutting bad rate **8.71% to 5.66%**; built PSI and calibration **monitoring** with 243 tests.

Sentiment Detection & Text Generation – *Deep learning: TensorFlow to PyTorch rebuild* GitHub

- Rebuilt a TensorFlow **LSTM** project in **PyTorch** as a FastAPI and Streamlit app with **444 tests**.
- Trained an LSTM **NLP** sentiment classifier (**macro-F1 0.830**); temperature calibration cut ECE 0.080 to 0.022.
- Fixed 11 defects, incl. a sampler drawing uniformly at every temperature and stopwords stripping all 14 negations.

MEDBOT – Medical RAG Chatbot – *Vector database, embeddings & prompt engineering* GitHub

- Built a **RAG** chatbot (LangChain, Gemini API, local embeddings) over a 1,225-chunk **FAISS** vector index, with citations.
- **Prompt-engineered** away false refusals, **18/46 to 0/46** questions, while holding a 120/120 hallucination guard.
- Cut prompt cost to 1/14th of the chain-of-thought version (216 vs 3,037 tokens); 175 offline tests and CI on every push.

Skills

Gen AI / LLM, Agentic AI & Prompt Engineering: LLMs (OpenAI, Google Gemini, Llama via Ollama), **Generative AI**, **RAG**, **Embeddings**, **AI Agents** (Tool Calling), **Prompt Engineering**, **LangChain** (LCEL), **Vector Databases** (FAISS, Qdrant), Hugging Face Transformers, LLM Evaluation (Golden Sets, RAGAS), LLM Fine-Tuning (LoRA/QLoRA), LLM Tracing (Langfuse), AI Guardrails

Model Lifecycle, MLOps & Deployment: Cloud (GCP Cloud Run; Azure deploy scripts), **Model Deployment**, **REST APIs** (FastAPI), **Git/GitHub**, **Model Monitoring** (PSI Drift), Docker, CI/CD (GitHub Actions), MLflow, pytest

Machine Learning & Predictive Modelling: Machine Learning, **Segmentation**, **Feature Engineering**, **Time-Series Forecasting**, **Fraud & Credit Risk**, **Tree-Based Models** (XGBoost, LightGBM, Random Forest), **Regression** (Linear, Logistic), Classification, Clustering, Graph Analytics (NetworkX), Recommendation Systems, Anomaly Detection, Model Evaluation, Cross-Validation

NLP & Deep Learning: NLP, **Deep Learning**, **PyTorch**, **TensorFlow**/Keras, Transformers, LSTM, Text Classification, Sentiment Analysis, NLTK, spaCy

Data Analysis & Statistics: Statistics, **A/B Testing & Experiment Design** (Power, MDE), **Funnel Analysis**, EDA, Hypothesis Testing, Causal Inference

Model Governance & Responsible AI: Explainability (SHAP), Model Validation, Leakage Detection

Databases & Data Engineering: SQL (CTEs, Window Functions, Joins), **Data Pipelines** (ETL, Prefect), Data Cleaning, PostgreSQL, DuckDB, Large Datasets (8M+ rows)

Languages, Libraries & Frameworks: Python, **SQL**, **R**, **NumPy**, **Pandas**, **Scikit-learn**, Statsmodels, Matplotlib, Seaborn

Visualization & Collaboration: Stakeholder Communication, **Excel**, PowerPoint, Tableau, Power BI, Technical Documentation, Streamlit

Certifications

Databases and SQL for Data Science with Python, IBM (Coursera)	Jan 2024
Machine Learning with Python, IBM (Coursera)	Dec 2023
Exploratory Data Analysis for Machine Learning, IBM (Coursera)	Dec 2023
Data Analysis with Python, IBM (Coursera)	Oct 2023
Python Packages for Data Science, University of Colorado Boulder (Coursera)	Oct 2023

Education

BSc Mathematics, Amrita Vishwa Vidyapeetham, Coimbatore	2017 – 2020
Data Science Bootcamp, Brototype, Bengaluru	2023 – 2024